Qiqi Liu

(Official Name on Transcripts: Jiji Liu)

E-mail: vrmeies@gmail.com | Phone: +86 18637312585 | Country: China | Gender: Male

Master's Graduate in CS | *Research: Trustworthy AI, LLM Safety, Multi-Agent System Security*

RESEARCH INTERESTS

My research investigates the safety, robustness, and trustworthiness of LLM-integrated systems, with a focus on multi-agent architectures and adversarial evaluation. I am particularly interested in characterizing attacks along a continuous syntactic–semantic spectrum and developing **adaptive defenses that respond to the interpretation depth required by each threat**. My extensive engineering background in generative AI (diffusion models, video synthesis, LoRA fine-tuning) provides me with a deep, systems-level understanding of the models I aim to secure.

EDUCATION

University of Chinese Academy of Sciences (UCAS), Beijing — 2022–2025

M.Sc. in Electronic Information (Recommended Admission) **| GPA: 3.64/4.00 | Ranking: Top 15%**

Henan University, Kaifeng — 2018–2022

Bachelor's Degree in Information Assurance **| Ranking: Top 5%**

PUBLICATIONS

Q. Liu, T. Holz, R. Song, S. Ye, “The Capability Paradox: How Smarter Auditors Make Multi-Agent Systems Less Secure,” Submitted to NeurIPS 2026 (LLM Robustness & Multi-Agent Security) [Under Review].

Contribution: Lead & corresponding author; proposed the capability paradox framework; responsible for experimental design, evaluation, and manuscript writing.

Q. Liu, Y. Sun, R. Song, H. Zhang, L. Sun, “MIDS: Monitoring Anomalies with Bidirectional Mamba,” Submitted to IEEE TIFS (Information Assurance & Systems Research) [Under Revision].

Contribution: Lead author; responsible for system architecture design, robustness evaluation within consumer-grade mobility scenarios, and manuscript writing.

R. Song, **Q. Liu**, J. Yin, L. Meng, L. Cui, W. Wang, Z. Ding, L. Li, Z. Hao, “MGCL: Tor Website Fingerprinting For Reload Scenario,” Submitted to Computer Networks (Elsevier) [Under Review].

Contribution: 2nd author; contributed to model design and experimental evaluation.

L. Zhang, Y. Huo, Q. Ge, Y. Ma, **Q. Liu**, W. Ouyang, “A Privacy Protection Scheme for IoT Big Data Based on Time and Frequency Limitation,” Wireless Communications and Mobile Computing, 2021. doi: 10.1155/2021/5545648.

Contribution: 2nd student author; privacy-mechanism validation, data integrity evaluation, and manuscript preparation.

RESEARCH EXPERIENCE

Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany (Remote) — 2026.2 – Present

Research Intern | *Advisor: Prof. Thorsten Holz*

- Discovered and formalized the “Capability Paradox” in hierarchical multi-agent systems—demonstrating that stronger auditor LLMs can paradoxically reduce overall system security by expanding attack surfaces through cross-context instruction handling.
- Designed systematic adversarial evaluation frameworks targeting tool-integrated agents; assessed semantic filter effectiveness and privilege isolation constraints across commercial coding agents (Roo Code, Cursor), identifying critical boundary edge-cases in production environments.
- First-authored a paper submitted to NeurIPS 2026 with Prof. Holz as co-author, proposing a new threat model for multi-agent security that challenges the assumption that more capable components yield safer systems.

China Telecom AI Research Institute (TeleAI), EVOL Lab — 2024.10 – 2025.4

AIGC Research Intern

- Designed a fidelity-preserving video synthesis model architecture based on depth map, incorporating reference and structural images to enhance synthesis stability.
- Built datasets using SAM2 and DAM; developed a four-stage training strategy and data augmentation method.
- Integrated ControlNeXt into CogVideoX, optimizing large-scale performance on 720K video clips (~7TB) with distributed training on 4×8×H100 GPUs.

Alibaba Group, AI Algorithm Department — 2024.6 – 2024.10

AIGC Research Intern

- Developed style image synthesis algorithms and fine-tuned multimodal models (SDXL, DiT); improved image fidelity metrics (Top-1 ↑20%).
- Gained hands-on experience with advanced samplers (DDPM, DDIM, ODE, flow-matching) and enhanced synthesis stability via LoRA on 5K image–text pairs.

Meizu Technology Co., AI Vision Algorithm Department — 2024.3 – 2024.6
AI Algorithm Intern

- Built a robust intelligent agent for assistive phone UI understanding; pretrained and fine-tuned (LoRA) vision-language models (Qwen-VL-Chat, GLM4V) on A100 GPUs with 50K annotated screenshots (~2M labels).
- Designed a verifiable OCR + detection + LLM framework (Damo OCR, GroundingDINO, GPT-4o) and transitioned to high-fidelity end-to-end multimodal models.

INDUSTRY EXPERIENCE

Zseads Technology Co., AIGC Algorithm Engineer — 2025.4 – 2025.12 (Intern → Full-time)

- Benchmarked quality and computational cost across multiple SOTA methods.
- Built a Flux-Fill restoration LoRA training framework, reducing memory usage to 25% of the original and adapting it for optimization strategies like ZeRO-3.
- Developed a high-quality hand restoration pipeline based on Wan2.1, achieving SOTA-level results.

TECHNICAL SKILLS

AI Safety & Robustness: System Resilience, Trustworthy AI, Adversarial Evaluation, Multi-Agent Security
Generative AI & LLM: Diffusion Models (SDXL, DiT), LoRA Fine-tuning, Flow-matching, Flux-Fill, CogVideoX
Tools & Frameworks: PyTorch, DeepSpeed (ZeRO-3), TensorRT, LangChain

PROFESSIONAL SERVICE

Reviewer, IEEE TIFS (Transactions on Information Forensics and Security), 2024–2025 (2 papers)
Reviewer, IEEE TCSVT (Transactions on Circuits and Systems for Video Technology), 2024–2025 (2 papers)
Reviewer, IEEE TMM (Transactions on Multimedia), 2024–2025 (1 paper)

SELECTED PROJECTS & COMPETITIONS

Diffusion-Based Long-term Time Series Synthesis (Project 3DTS)

- Proposed a novel diffusion probabilistic framework for time-series modeling to address the scalability bottleneck of Transformer-based models over extended horizons.
- Designed a custom refinement network tailored for temporal stability and conducted scalability analysis comparing with SOTA baselines.

Selected Competitions:

- 2023, 8th Place Nationwide, DataCon Data Integrity and Reliability Competition
- 2019, First Prize, ISCC National Information Protection and Technical Excellence Competition

LANGUAGES

English: CEFR B2 (Independent User) **| Chinese:** Native

REFERENCES

Prof. Thorsten Holz | Faculty, Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany
Relationship: Research advisor (2026.2–Present) **|** E-mail: available upon request
Additional references available upon request.